NLP - Assignment 4

Group 4

Task 1

- DistilBERT is a lighter, faster version of BERT
- Created using knowledge distillation
- A smaller model learns from a larger pre-trained BERT
- More efficient in memory and computation

Inputs and Outputs

- Input: raw text
- Text → tokens → numerical form:
  - input IDs
  - attention masks
- Key difference from BERT:
  - No token type IDs (simplified architecture)
- Output: logits (scores)
- After normalization → probabilities
- Final prediction = class with highest probability

Number of Classes &Input Size

- Two classes: Positive/Negative
- Maximum sequence length: 512 tokens
- In our setup: 256 tokens
- Short inputs → padded
- Long inputs → truncated
- Trade-off:
  - Shorter length = faster training
  - Longer length = more context

Case Sensitivity

- We used uncased version
- Advantages:
  - More robust to text variations
- Limitation:
  - Loses information from capitalization

Use for Azerbaijani

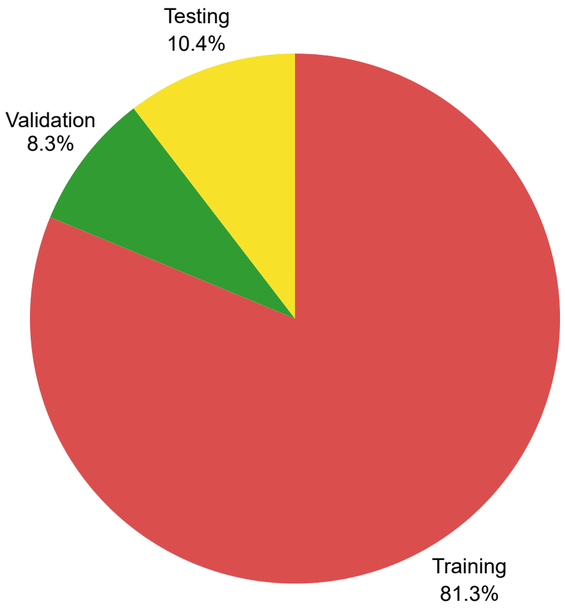
- Pretrained mainly on English data
- Azerbaijani is agglutinative:
  - Words formed with many suffixes
- English-trained model struggles with this structure
- Works, but not optimal
- Better alternatives:
  - Multilingual models
  - Language-specific models

Results

- IMDB English dataset
- Accuracy: 91%
- Example: "This is a really good movie. I loved it and will watch again"
    - Prediction: Positive
    - Probabilities: [[0.00724872 0.99275124]]

Squad 1.1

Training	Validation	Testing
78k	8k	10k



BiDAF Model

I.		II.		III.		IV.		V.
Embedding	→	Contextual Encoding	→	Attention Flow	→	Modeling	→	Output

BiDAF Model

Embedding → Contextual Encoding →

$$x_i = Embedding(w_i)$$

GloVe

BERT

$$h_i = [\overrightarrow{hi}; \overleftarrow{hi}]$$

BiLSTM

BiDAF Model

→ Attention Flow → Modeling →

$$S_{ij} = w\top[c_i; q_j; c_i \circ q_j]$$

M=BiLSTM(G)

BiDAF Model

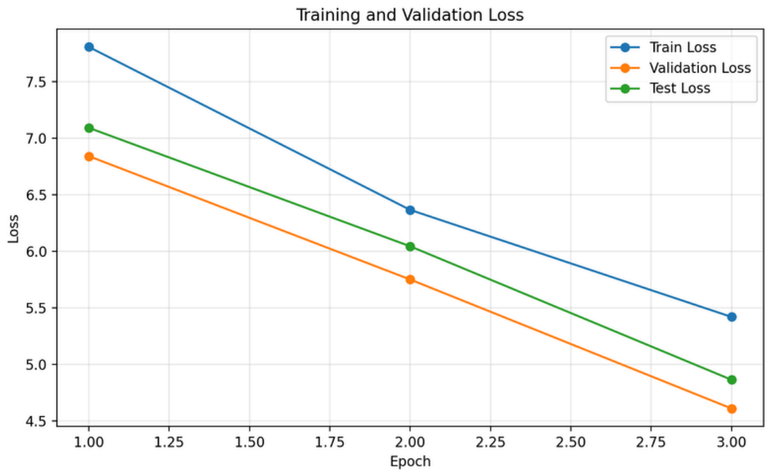
→ Output

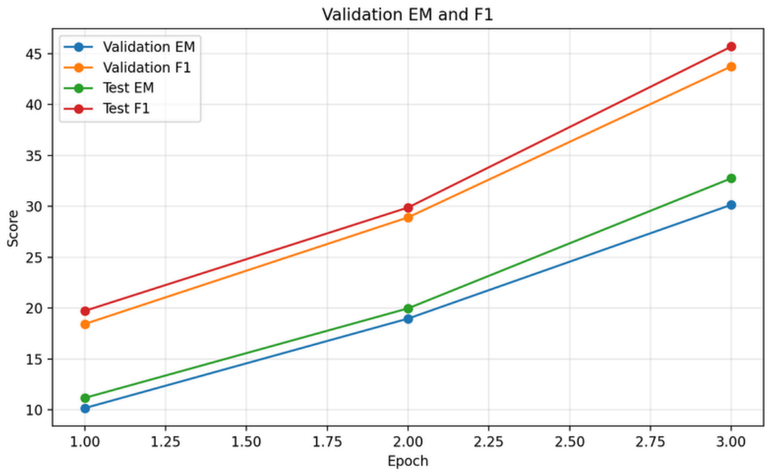
$$p^{start} = softmax(W_s[G; M])$$

$$p^{end} = softmax(W_e[G; M_2])$$

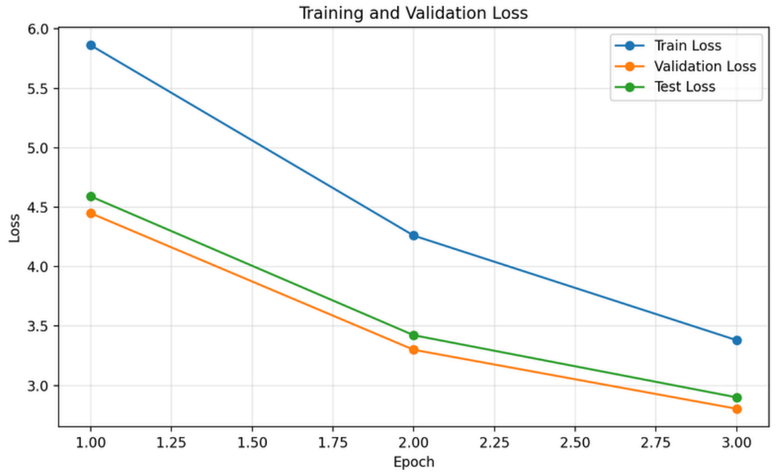
Training Process

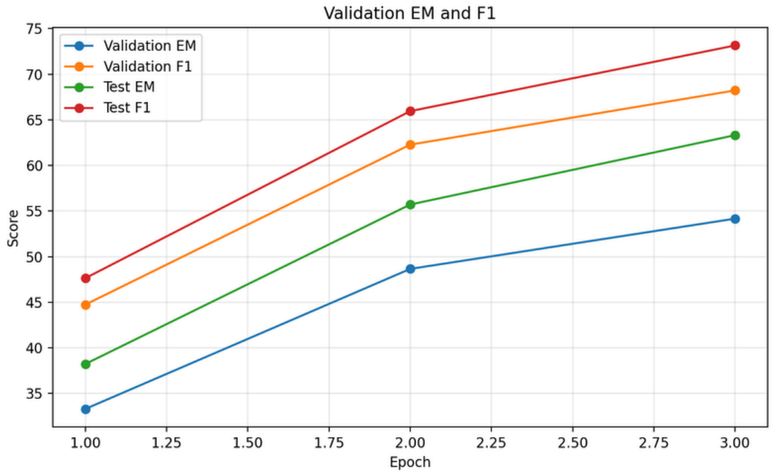
GloVe





BERT





UI

Try a prediction

Use a preset example or type your own question and context.

Model
bert_bidaf

Sample questions

Click to fill the fields

Capital city
What is the capital of Azerbaijan?

Currency
What currency is used in Azerbaijan?

Historic landmark
In which city is the Maiden Tower located?

Question

What is the capital of Azerbaijan?

Context

Azerbaijan is a country in the South Caucasus. Its capital is Baku, which lies on the western coast of the Caspian Sea.

Max length

30

Get answer

Result

Run a prediction to see the extracted answer and span.